

7ai	Any one of the followings: Does not leak / longer cycle life / faster charging / higher degree of freedom in shape / higher current density / less flammable.	A	A1
7aii1	Any one of the followings: Electric vehicle / personal mobility device / car battery / electric scooter / solid-state solar cell or panel.	A	A1
7aii2	Any one of the followings: Pacemakers / radio frequency identification (RFID) / wearable devices / handphone or smart phone / smart card / implantable medical device e.g. implantable defibrillator / wireless sensor / remote control / calculator.	A	A1
7aiii	Any one of the followings: 1. Bulk solid-state battery can store more energy than thin film solid-state battery. 2. Thin film solid-state batteries have longer cycle life than bulk solid-state battery. 3. Thin film solid-state battery can be easily shaped based on application while bulk-solid state has more standard construction shape.	A	B1
7bi	Current per unit area	E	B1
7bii	Energy per unit mass	E	B1
7ci	Charge storage capacity = charge capacity × area = 3.6×50 = 180 C	E	A1
7cii	Recommended max. discharge current = recommended max. discharge current density × area = 0.15×50 = 7.5 mA	A	A1
7ciiii	$Q = It$ $180 = (7.5 \times 10^{-3})t$ $t = 24\,000\text{ s}$	A	C1 A1
7civ	$E = QV$ = $(180)(2.5)$ = 450 J Alternatively. $E = VIt = (2.5)(7.5 \times 10^{-3})(24000)$ = 450 J	A	C1 A1
7di	<u>Total charge supplied</u> to the cell	A	B1
7dii	Estimate area under graph, then use the area divided by total time 9 mA. Acceptable range: 8 – 10 mA, answer must be 1 – 2 sf.	A	A1
7e	Efficiency = (discharged energy / energy used to charge cell) × 100% = $(450 / 550) \times 100\%$ = 82%	A	C1 A1
7fi	Above 140 °C, cells may overheat and catch fire / result in burn hazard OR	A	B1

	Plastic film / lithium metal anode may melt.		
7fii	Water or water vapour within lithium cells may result in electric shocks / short circuit. OR Lithium reacts violently with water (producing a strong base that is harmful to human body / producing highly flammable hydrogen gas)	A	B1
7g	Since each cell has average e.m.f. of 2.5 V. Number of cells = $10 / 2.5 = 4$ <u>4 cells arranged in series</u>  Since recommended max current density of 0.15 mA per cm² Electrode area = $300 / 0.15 = \underline{2000 \text{ cm}^2}$	D	B1 B1